

1 Introduction

Ethernet is available in different speeds (10/100/1000 and 10000Mbps) and provides connectivity to meet a wide range of needs from desktop to switches. MorethanIP IP solutions provide a solution for each Ethernet application with a library of configurable MAC (Media Access Control) and PCS (Physical Coding Sub-layer) Cores.

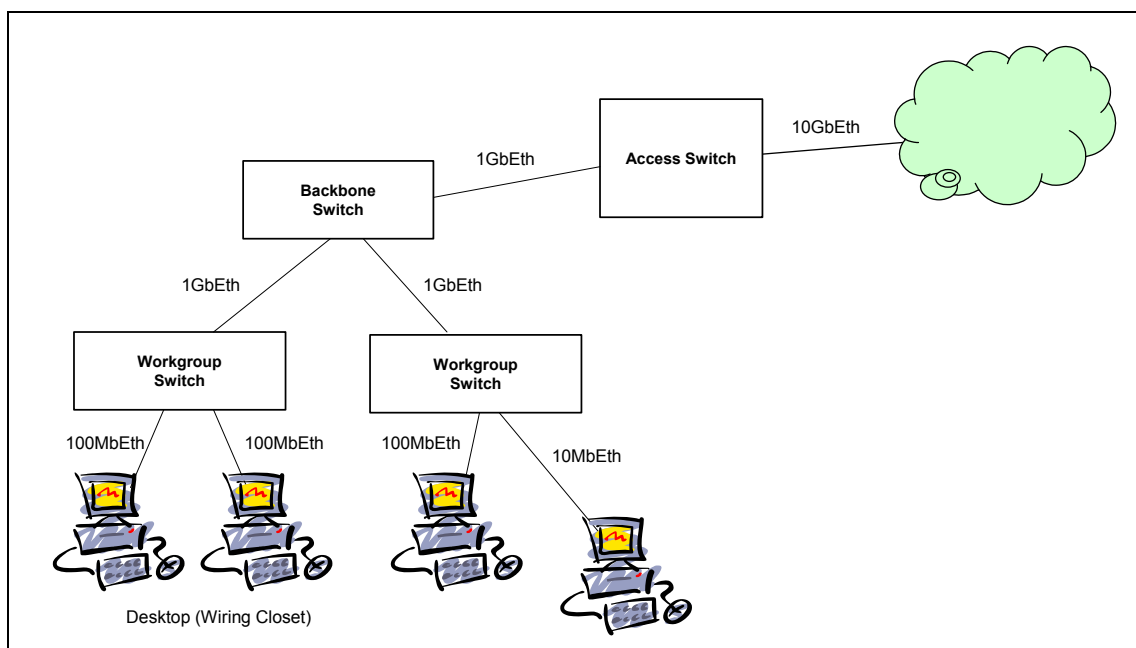


Figure 1: Enterprise LAN Topology Example

The programmable 10/100/100 Ethernet MAC from MorethanIP provides, with a single IP Core, a solution for Ethernet applications (Line Card, NIC card or switching) operating at 10/100 or 1000Mbps (Gigabit Ethernet). The 10/100/1000 MAC Core operates Full Duplex mode, supports transparent (For switching applications) and full Ethernet frame termination / generation (For NIC or line cards applications).

For efficient power management, the Core also implements Magic Packets detection.

The core can seamlessly connect to any industry standard Gigabit Ethernet PHY device via a Gigabit Medium Independent Interface (GMII for 1000Mbps application) or Medium Independent Interface (MII for 10/100Mbps applications) and to a user application via a SOC (System on a Chip) interface which provides seamless connectivity to any MorethanIP cores such as Flexbus, POS-PHY, PCI interfaces or any third party Core which implements an interface compatible with the Altera Atlantic specification.

2 Application Example

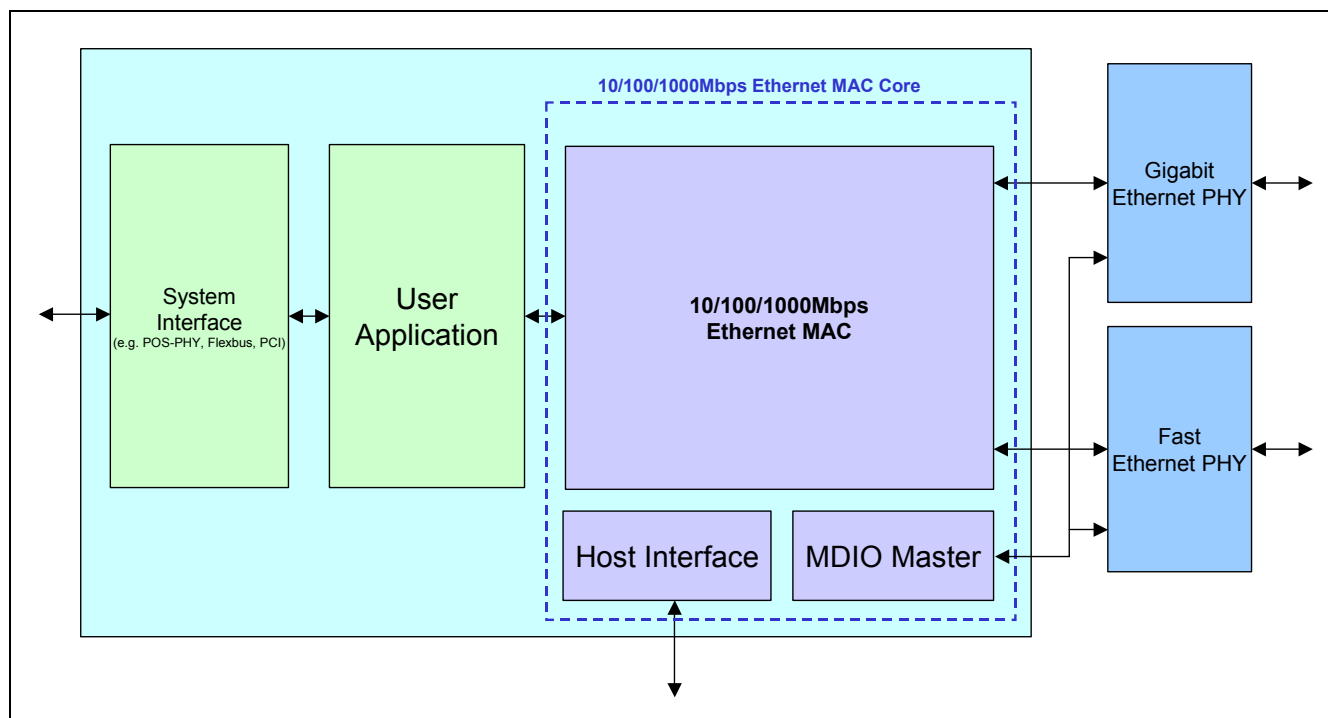


Figure 2: Application Example

3 10/100/1000Mbps Ethernet MAC core Features

- Implements the full 802.3 specification with preamble / SFD generation, frame padding generation, CRC generation and checking on transmit and receive respectively
- Dynamically configurable to support 10Mbps, 100Mbps or 1Gbps operation
- Supports AMD Magic Packet detection for node remote power management
- Supports, for 10/100Mbps operation, full duplex or half duplex operation selectable via a Core configuration option
- Seamless interface to commercial Gigabit Ethernet PHY device via a 8-Bit Gigabit Medium Independent Interface (GMII) operating at 125MHz
- Seamless interface to commercial Fast Ethernet PHY device via a 4-Bit Medium Independent Interface (MII) operating at 25MHz
- Simple FIFO interface to user application compatible with Altera Atlantic SOC (System On Chip) interface
- CRC-32 checking at full speed using a multi-stage CRC calculation architecture with optional forwarding of the FCS field to the user application
- CRC-32 generation and append on transmit or forwarding of user application provided FCS selectable on a per-frame basis
- When operating in Full Duplex mode, implements fully automated XON and XOFF Pause Frame (802.3 Annex 31A) generation and termination providing flow control without user application intervention
- When operating in full duplex mode, pause quanta used to form XOFF Pause frames, dynamically programmable

- Pause frame generation additionally controllable by user application offering flexible traffic flow control
- Optional forwarding of received pause frames to the user application when operating in Full Duplex mode
- Implements standard flow-control mechanism in full-duplex operation mode
- In half-duplex mode, provides full collision support, including jamming, backoff, and automatic retransmission
- Support for VLAN tagged frames according to IEEE 802.1Q
- Programmable MAC address: Insertion on transmit; discards frames with mismatching destination address on receive (Except Broadcast and frames)
- Programmable group of four supplemental MAC addresses that can be used to accept / reject Unicast traffic
- Programmable Promiscuous mode support to omit MAC destination address checking on receive
- Multicast address filtering on receive based on 64 entries hash table reducing higher layer processing load
- Programmable frame maximum length providing support for any standard or proprietary frame length (e.g. 9K-Bytes Jumbo Frames)
- Optional internal GMII / MII Loopback
- Statistics indicators for frame traffic as well as errors (alignment, CRC, length) and pause frames providing for IEEE 802.3 basic and mandatory Management Information Database (MIB) package as well as Ethernet MIB (RFC 2665) and Remote Network Monitoring (RFC 2819) enabling implementation in SNMP management environments
- Simple handshake user application FIFO interface with fully programmable depth and threshold levels ensuring data rates of 1Gbps with full back-to-back frame transfer support
- 8-Bit or 32-Bit Client FIFO interface
- Optional automatic receive errored frame discard
- Separate status word available for each received frame on the user interface providing information such as frame length, frame type, VLAN tag and error information
- MDIO Master interface for PHY device configuration and management with two programmable MDIO base addresses
- Available for FPGA or ASIC implementation
- Design Kit comes with extensive Ethernet frame generators and checking models enabling fully automated design verification and testing for standard compliance and error behavior, enabling for fast turn-around design cycles

4 MAC Core Block Diagram

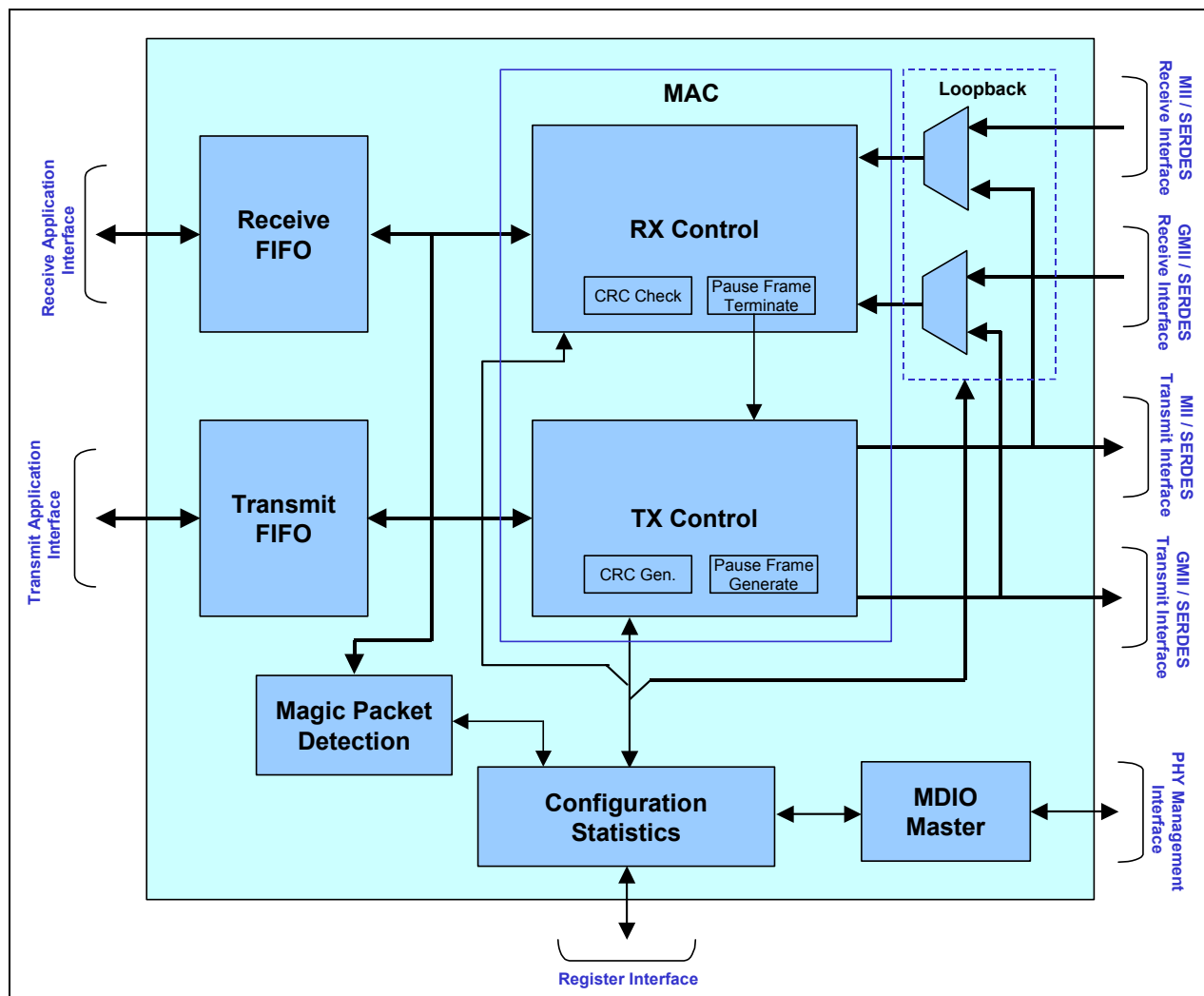


Figure 3: 10/100/1000Mbps Ethernet MAC Core Overview

5 Implementation Summary

Table 1: 10/100/1000 Ethernet MAC complexity Summary

Target Device Family	Speed Grade	Complexity (With 256 Bytes FIFOs)	Performance	Requirement (Gigabit Ethernet)
STRATIX II	-5	3400 LEs	170MHz	125MHz
CYCLONE II	-8	3600 LEs	140MHz	125MHz
STRATIX	-7	3600 LEs	140MHz	125MHz
CYCLONE	-8	3600 LEs	130MHz	125MHz

6 10/100/1000Mbps Ethernet MAC Design Kit Overview

Table 2: Design Kit Overview

<i>Design and Simulation</i>	
Language	Optimized VHDL / Verilog or lower cost FPGA encrypted netlist.
Simulation	Configurable VHDL / Verilog Testbench with embedded frame generator and checker providing an easy to use and robust de-bugging environment.
Verification	Comprehensive test environment with Ethernet frame generator and verification models for standard compliant and errored frame generation and automated core behavior verification.
<i>Supported Design Tools</i>	
Simulation	Modelsim Version 5.7a or higher.
Synthesis	Quartus II 4.1 or higher
Implementation	Quartus II 4.1r higher

7 References

1. IEEE 802.3 2002 Edition
2. IEEE 802.1Q 1998 Edition
3. RFC2665, Definitions of Managed Objects for the Ethernet-like Interface Type, www.ietf.org
4. RFC2863, The interfaces Group MIB, www.ietf.org
5. AMD, Magic Packet Technology White Paper, Publication 20213 Rev. A November 1995

8 Ordering Code

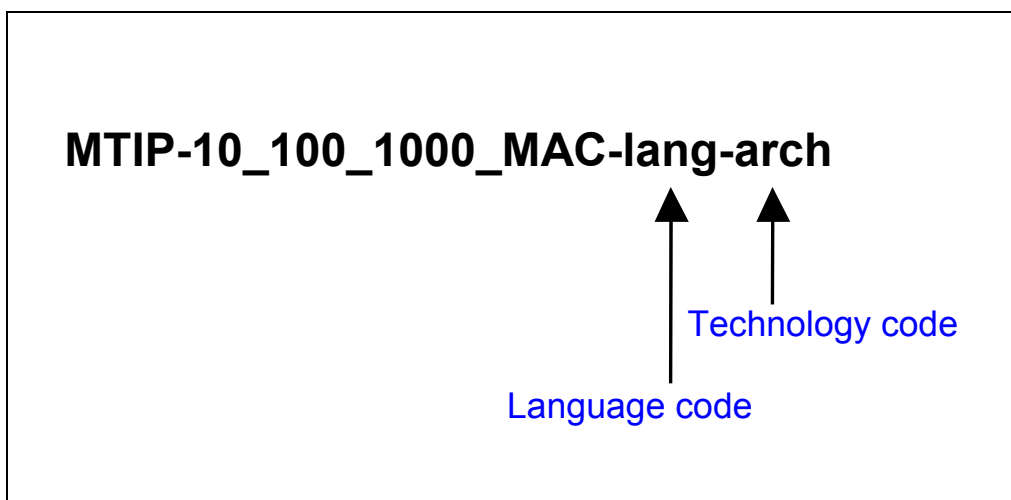


Table 3: Language Code

Technology Code	Target Technology
BIN	Encrypted FPGA netlist.
VHDL	Synthesizable generic VHDL source code for FPGA or ASIC implementations
VLOG	Synthesizable generic Verilog source code for FPGA or ASIC implementations

Table 4: Technology Code

Technology Code	Target Technology
GEN	Source code option for Altera FPGAs (CYCLONE, CYCLONE –II, STRATIX, STRATIX-II or STRATIX GX) or ASIC implementations.
ALTR	Encrypted netlist for Altera FPGAs (CYCLONE, CYCLONE –II, STRATIX, STRATIX-II or STRATIX GX).

9 Contact

MorethanIP

E-Mail : info@morethanip.com

Internet : www.morethanip.com

Europe

An der Steinernen Bruecke 1

D-85757 Karlsfeld

Germany

Tel : +49 (0) 8131 333939 0

FAX : +49 (0) 8131 333939 1